

Programming work: Secure web app with user login and registration

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General

- Create a web application with register and login features
- Main page is only accessible after logging in
- Secure practices
- Not based on earlier work
- SPA

Technologies

- Frontend
 - Node, React, JS
 - Axios, React Router, Redux, Formik, Yup
- Backend
 - Express, SQLite3, JS
 - Bcrypt, cookie-parser, cors, dotenv, express & -rate-limit & -session, helmet, jsonwebtoken, zod, crypto

Secure programming aspects

- Register form
 - Input validation
 - Parametrized query for database actions
- Login
 - Bruteforce protected
- XSS and CSRF
 - Output encoding,
 - CSRF tokens, manual STP
- Access + refresh tokens
 - JWT sameSite, httpOnly
- Safe error messages (not leaking email or username info)
- Password hashed and salted for storing

Testing

- Work in progress
- OWASP offers testing information and snippets
- Unit testing, integration testing & manual testing at least

AI usage

- Planning, building overall picture
- Reviewing full files
- Debugging